Project Budget – Minimum Viable Product (MVP) Prototype

The proposed budget supports development of a **flight-testable Minimum Viable Product (MVP)** of the Modular Operational Network Kernel (MONK) governance payload within the University of Maryland UAS Research and Operations Center internship material allocation of approximately **$1,000**.

The budget has been intentionally structured to prioritize mission-critical hardware components while maintaining strict adherence to the program's material allocation limits, ensuring responsible use of research resources and feasibility within the 10-week internship schedule.

The drone airframe and flight control platform will be provided through UROC flight testing resources associated with the Chesapeake UAS Route Network (CURN). This budget therefore focuses exclusively on the modular hardware components required to implement and test the MONK telemetry governance payload.

Budget Breakdown

Item Category	Technical Component	Project Justification	Estimated Cost	Running Total
Edge Compute	Raspberry Pi 5 (8GB) + NVMe Base + UPS Power HAT	Edge processor hosting the MONK Governance Kernel. UPS module ensures reliable telemetry logging during power interruptions or link-loss events.	$175	$175
PNT Sensor	SparkFun u-blox ZED-F9P RTK GNSS Module	Provides high-precision Positioning, Navigation, and Timing (PNT) data for validating corridor compliance and geofence verification.	$275	$450
Secure Identity	TPM 2.0 SPI Security Module	Hardware trust anchor used for cryptographic signing of telemetry records and generation of MONK-Key compliance tokens.	$35	$485
Connectivity	Sixfab LTE / 5G Cellular HAT	Enables telemetry broadcast capability for transmitting compliance evidence to monitoring or corridor governance systems.	$250	$735
Interface	MAVLink UART Interface Cables and TTL-USB Adapters	Communication bridge between the MONK payload and the drone flight controller using MAVLink telemetry.	$65	$800

Item Category	Technical Component	Project Justification	Estimated Cost	Running Total
Sensor Validation	BNO085 Absolute Orientation Sensor (9-axis IMU)	Independent motion sensing for anomaly detection and cross-validation of flight telemetry data.	$20	$820
Integration	Modular Quick-Release Payload Chassis	Lightweight payload mount fabricated using UROC 3D printing resources for safe integration with the drone airframe.	$110	$930
Development Environment	Edge runtime and telemetry logging environment	Supports software development, telemetry recording, and system testing during prototype development.	$70	$1,000
TOTAL PROJECT COST			**$1,000**	**$1,000**

Budget Justification

This budget supports development of a **flight-testable governance payload prototype** capable of validating drone telemetry and producing cryptographically verifiable compliance records during experimental flight operations.

The proposed components enable the MONK module to:

- ingest real-time MAVLink telemetry from the flight controller
- verify mission corridor compliance and geofence constraints
- detect flight anomalies using independent sensor validation
- generate tamper-resistant cryptographic telemetry records using a Trusted Platform Module
- produce a signed compliance token (**MONK-Key**) that can be used for corridor governance or future UAS Traffic Management (UTM) systems

The selected components allow the system to be assembled, integrated, and flight tested within the **10-week internship timeframe** while remaining within the **$1,000 material allocation**.

The component selection and cost structure reflect a deliberately scoped **Minimum Viable Product (MVP)** development strategy aligned with **Project Management Institute (PMI) project delivery principles**, ensuring that the prototype can be engineered, integrated, and evaluated within the internship schedule and available resources.